

Persuasion in Ordered Posterior Mean

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Motivation

Scenario

Netflix promotes its subscription by disclosing information about all the included shows at the same time. After seeing this information, a consumer tries to identify the five best shows. If all of them are worth watching, he will subscribe.

- Multiple uncertain values are disclosed at the same time
- Only one of them is pivotal
- Which is pivotal is determined by the rank after disclosure
- Information also determines the expected value of the pivotal item

The joint determination of identity and value disclosure raises a major challenge in these multi-dimensional information design settings.

Model

State: a value profile $\mathbf{v} = (v_1, v_2, \dots, v_n)$

Prior Belief: $\mathbf{v} \sim F$

- F is exchangeable, i.e. for any permutation $\sigma: \mathbb{N} \rightarrow \mathbb{N}$,
 $F(\mathbf{v}) = F(\sigma\mathbf{v}) = F(v_{\sigma(1)}, v_{\sigma(2)}, \dots, v_{\sigma(n)})$.
- F has bounded support normalized to $V \subset [0, 1]^n$.

Signal: joint distribution $(\mathbf{v}, s) \sim \pi$ with marginal $\mathbf{v} \sim F$.

- For every s , π determines

$$\pi(\mathbf{v}|s) \longrightarrow \mathbf{m}(s) := \mathbb{E}[\mathbf{v}|s] \longrightarrow m_{(k)}(s)$$

where $m_{(k)}(s)$ is the k -th largest coordinate in $\mathbf{m}(s)$. Denote it $m(s)$.

Definition (Inducibility)

A distribution G is k -induced by a signal structure π if

$$m_{(k)}(s) \sim G.$$

A distribution G is k -inducible if it is k -induced by some signal structure.

Research Question

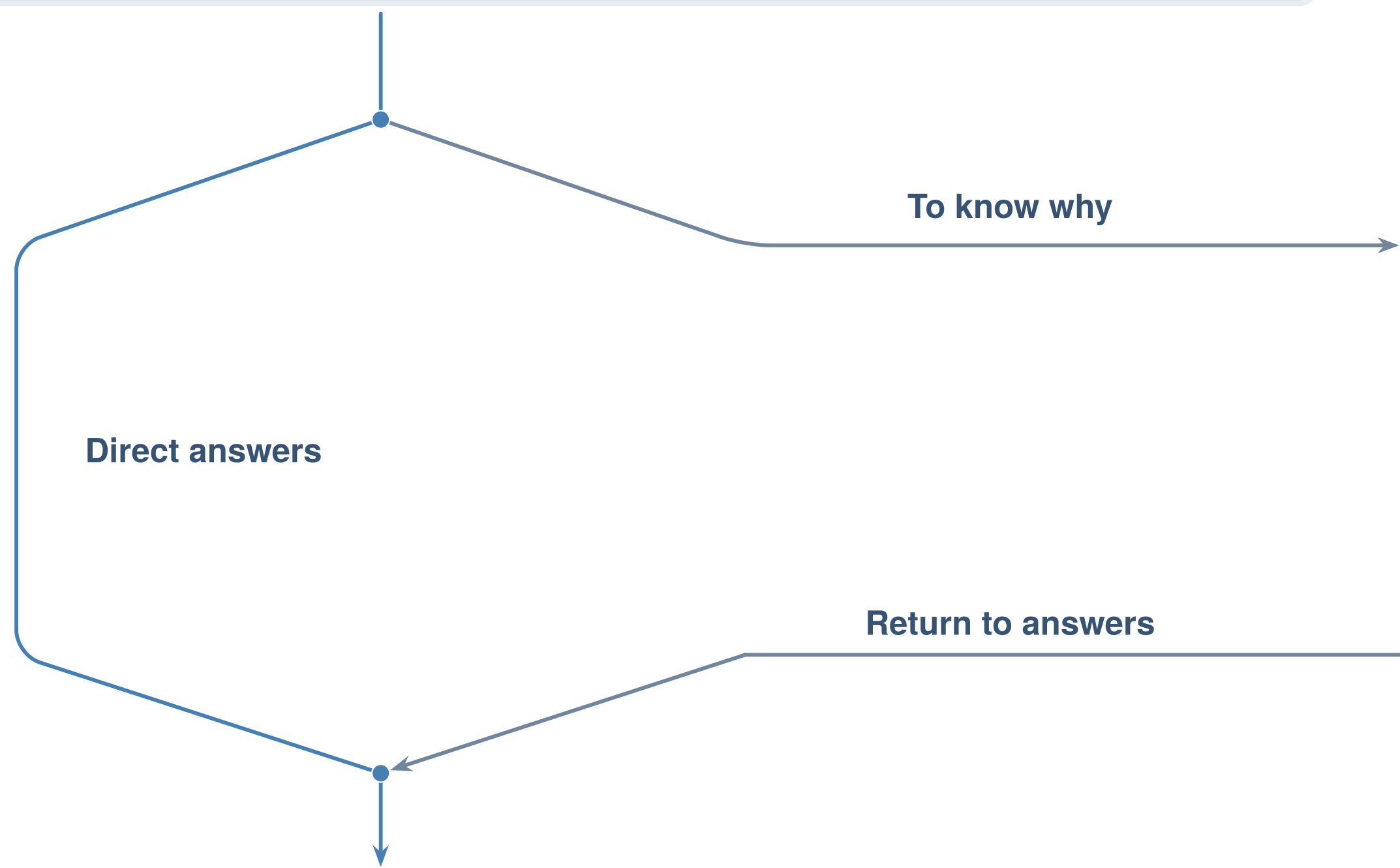
- Characterize the set of k -inducible distributions $\mathcal{G}$
- Given any functional $U: \Delta([0, 1]) \rightarrow \mathbb{R}$, make the following optimization problem tractable:

$$\max_{G \in \mathcal{G}} U(G).$$

Example (Scenario Revisited)

Let $\mathbf{v} = (v_1, v_2, \dots, v_n)$ denote the shows' values, where v_i is the value of show i . Netflix sends information about the shows. After receiving signal s , a consumer forms the posterior belief $\pi(\mathbf{v}|s)$ about the shows' values and computes the posterior-mean vector $\mathbf{m}(s)$. The consumer subscribes if and only if the fifth-largest posterior mean is at least p , i.e., $m_{(5)}(s) \geq p$. To maximize the probability of subscription, Netflix solves the following problem.

$$\max_{G \in \mathcal{G}} \int_0^1 \mathbb{1}\{m(s) \geq p\} dG(m)$$

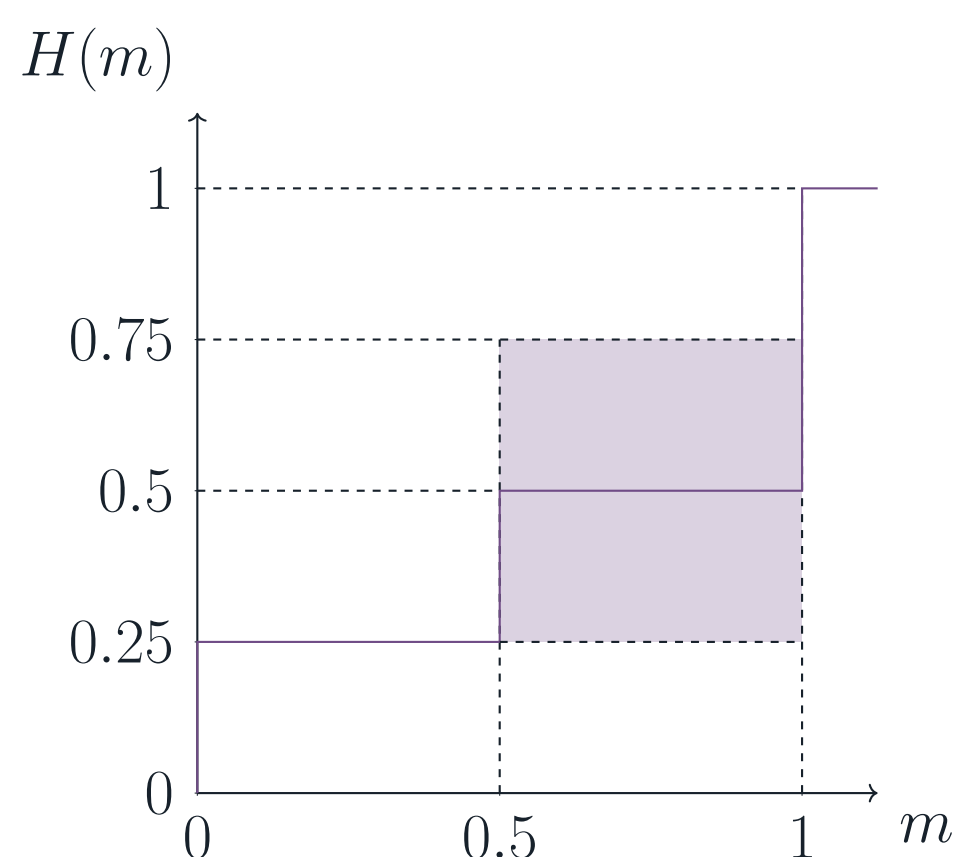


Binary Example

When $n = 2$, $v_1, v_2 \sim \text{i.i.d. Binary}(0.5)$, and $k = 1$, we have

$$\mathcal{G} = \bigcup_{H \in \mathcal{H}} \text{MPC}(H).$$

$\mathcal{H}$ is characterized by



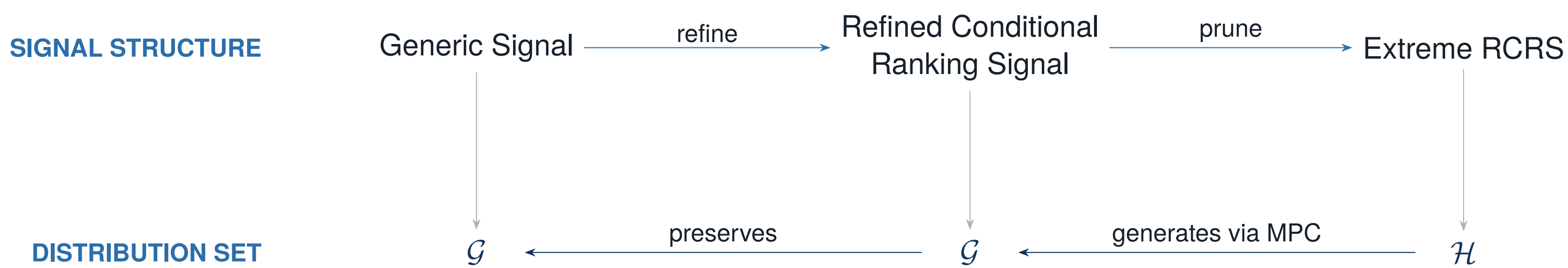
- Choose a horizontal segment within the shaded region.
- Join it to the fixed left and right portions to obtain $H \in \mathcal{H}$.
- Conversely, every $H \in \mathcal{H}$ arises from one such choice.

Implications under Properties of U

Properties of U	Optimal G in
$G_1 \succsim_{\text{FOSD}} G_2 \implies U(G_1) \geq U(G_2)$	$\text{MPC}(F_{\bar{v}^+})$
$G_1 \succsim_{\text{FOSD}} G_2 \implies U(G_1) \leq U(G_2)$	$\text{MPC}(F_{\bar{v}^-})$
$G_1 \succsim_{\text{SOSD}} G_2 \implies U(G_1) \geq U(G_2)$	$\{\delta_x\}_{x \in [\mathbb{E}[\bar{v}^-], \mathbb{E}[\bar{v}^+]]}$
$G_1 \succsim_{\text{SOSD}} G_2 \implies U(G_1) \leq U(G_2)$	$\mathcal{H}$

$$\bar{v}^+ := \frac{v_{(1)} + \dots + v_{(k)}}{k}, \quad \bar{v}^- := \frac{v_{(k)} + \dots + v_{(n)}}{n - k + 1}.$$

Roadmap



Information Decomposition

Definition (Components of Information)

For each realization $\mathbf{v}$, define its *identity-free value profile*

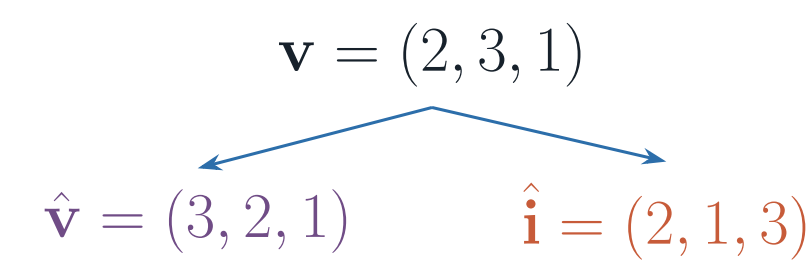
$$\hat{\mathbf{v}}(\mathbf{v}) = (v_{(1)}(\mathbf{v}), v_{(2)}(\mathbf{v}), \dots, v_{(n)}(\mathbf{v}))$$

where $v_{(\ell)}(\mathbf{v})$ is the value of the ℓ -th-largest coordinate of $\mathbf{v}$. Define its *ordering*

$$\hat{\mathbf{i}}(\mathbf{v}) = (i_{(1)}(\mathbf{v}), i_{(2)}(\mathbf{v}), \dots, i_{(n)}(\mathbf{v}))$$

where $i_{(\ell)}(\mathbf{v})$ is the identity of the ℓ -th-largest coordinate of $\mathbf{v}$.

Example

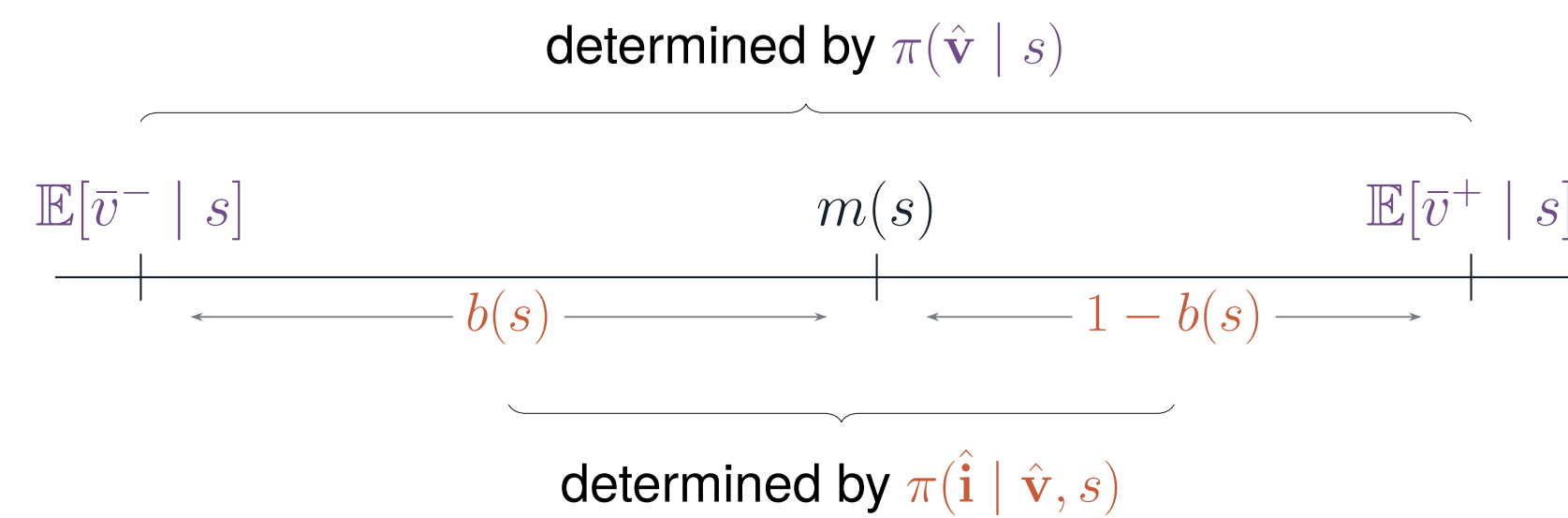


$$\begin{aligned} \text{Value Information} &= \text{Identity-free Value Information} + \text{Conditional Ranking Information} \\ \pi(\mathbf{v} | s) &= \pi(\hat{\mathbf{v}} | s) \times \pi(\hat{\mathbf{i}} | \hat{\mathbf{v}}, s) \end{aligned}$$

Information Refinement

$$\mathbb{E} \left[\underbrace{\frac{v_{(k)} + \dots + v_{(n)}}{n - k + 1}}_{:= \bar{v}^-(\hat{\mathbf{v}})} \middle| s \right] \leq m(s) \leq \mathbb{E} \left[\underbrace{\frac{v_{(1)} + \dots + v_{(k)}}{k}}_{:= \bar{v}^+(\hat{\mathbf{v}})} \middle| s \right].$$

- The two bounds are pinned down by $\pi(\hat{\mathbf{v}}|s)$ only
- Any $m(s)$ in the range can be achieved by choosing $\pi(\hat{\mathbf{i}} | \hat{\mathbf{v}}, s)$



Refined Conditional Ranking Signal

Definition (RCRS)

Parameters:

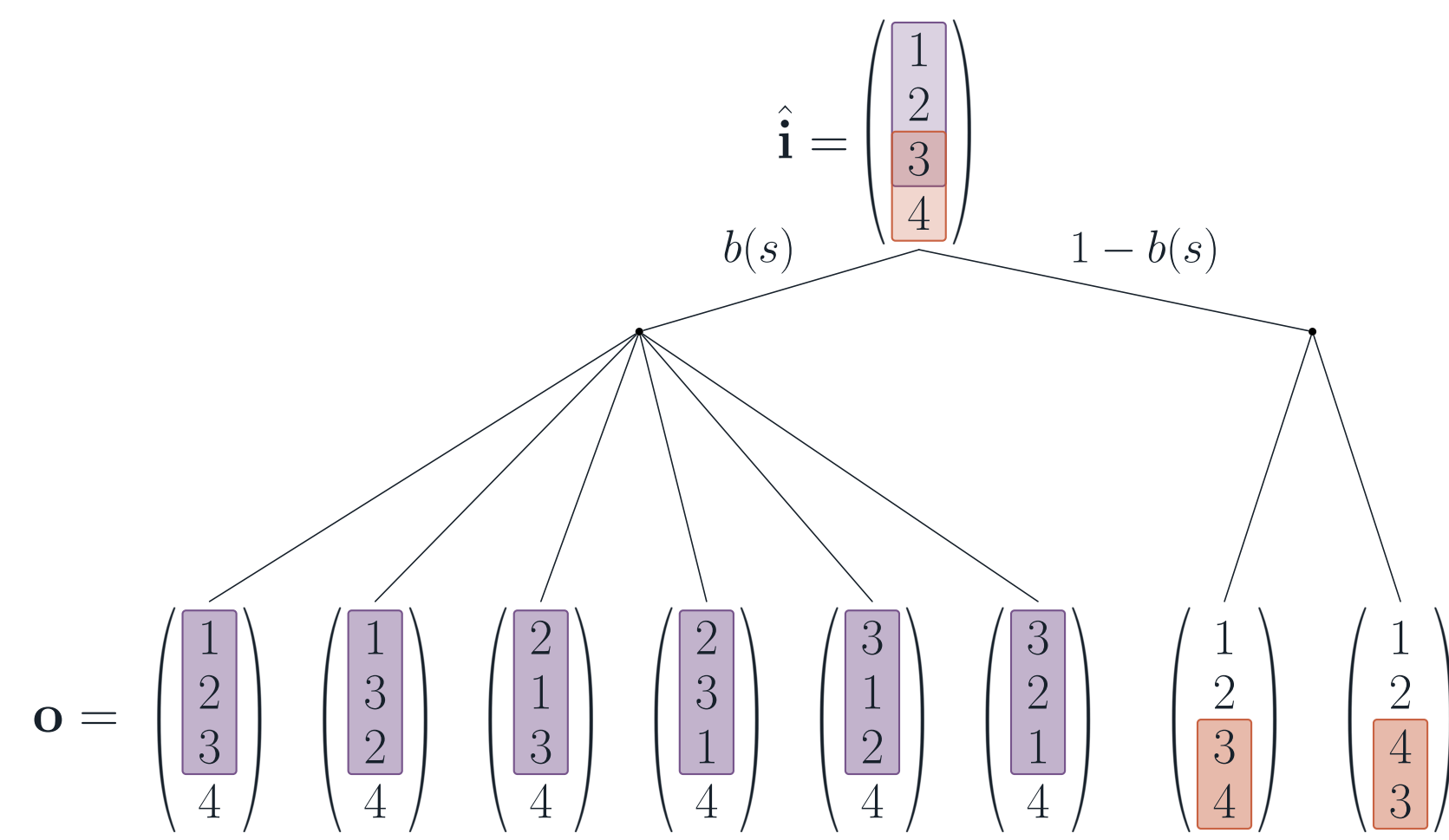
- ID-free kernel $q: [0, 1]^2 \rightarrow \Delta(S)$
- Ordering bias $b: S \rightarrow [0, 1]$

Draw a signal $(s, \mathbf{o})$ according to

- Given $\mathbf{v}$, determine s
 $s \sim q(\bar{v}^+, \bar{v}^-)$

$$\begin{aligned} \text{With probability } b(s) \quad \mathbf{o} &= \begin{pmatrix} \sigma \begin{pmatrix} i_{(1)} \\ \vdots \\ i_{(k-1)} \\ i_{(k)} \\ i_{(k+1)} \\ \vdots \\ i_{(n)} \end{pmatrix} \\ \end{pmatrix} \\ \text{With probability } 1 - b(s) \quad \mathbf{o} &= \begin{pmatrix} i_{(1)} \\ \vdots \\ i_{(k-1)} \\ \sigma \begin{pmatrix} i_{(k)} \\ i_{(k+1)} \\ \vdots \\ i_{(n)} \end{pmatrix} \end{pmatrix} \end{aligned}$$

Example (Order Signal Generation)



Theorem

Any k -inducible distribution can be k -induced by a refined conditional ranking signal.

Proof Sketch

Construct the ID-free kernel

$$q(\bar{v}_0^+, \bar{v}_0^-) = \pi(\cdot | \bar{v}_0^+, \bar{v}_0^-)$$

Construct the ordering bias

$$m^o(s) = b(s)\mathbb{E}[\bar{v}^+|s] + (1 - b(s))\mathbb{E}[\bar{v}^-|s].$$

Here, $m^o(s)$ denotes the expectation induced by the original signal structure, whereas the right-hand side is the expectation induced by the constructed signal structure.

Example (Kernel Construction)

$\backslash s$	s_1	s_2	$(\bar{v}^+, \bar{v}^-)$	s_1	s_2
$(\bar{v}_1^+, \bar{v}_1^-)$	$\mathbf{v}_1$	$\pi_{11} \quad \pi_{12}$	$(\bar{v}_1^+, \bar{v}_1^-)$	π_{11}	π_{12}
$(\bar{v}_2^+, \bar{v}_2^-)$	$\left\{ \begin{array}{l} \mathbf{v}_2 \\ \mathbf{v}_3 \end{array} \right.$	$\begin{array}{ c c } \hline \pi_{21} & \pi_{22} \\ \hline \pi_{31} & \pi_{32} \\ \hline \end{array}$	$(\bar{v}_2^+, \bar{v}_2^-)$	$\pi_{21} + \pi_{31}$	$\pi_{22} + \pi_{32}$
$(\bar{v}_3^+, \bar{v}_3^-)$	$\mathbf{v}_4$	$\pi_{41} \quad \pi_{42}$	$(\bar{v}_3^+, \bar{v}_3^-)$	π_{41}	π_{42}

(a) Original signal structure

(b) First-stage signal structure

Extreme Refined Conditional Ranking Signal

Only s matters for m

$$m(s, \mathbf{o}) = m(s, \mathbf{o}'), \forall s, \mathbf{o}, \mathbf{o}'$$

Information carried by s

$$\begin{aligned} \pi(\bar{v}^+, \bar{v}^- | s) & \quad \text{and} \quad b(s) \\ \uparrow \text{mix} & \quad \uparrow \text{mix} \\ \delta_{\bar{v}^+, \bar{v}^-} & \quad 0, 1 \end{aligned}$$

Definition (Extreme RCRS)

An RCRS is *extreme* if every s

- identifies $(\bar{v}^+, \bar{v}^-)$
- has $b(s) \in \{0, 1\}$

An extreme RCRS is pinned down by *revealed ordering bias*

$$\beta: [0, 1]^2 \rightarrow [0, 1]$$

$\beta(\bar{v}^+, \bar{v}^-)$ is the conditional probability that $b(s) = 1$, given that s identifies $(\bar{v}^+, \bar{v}^-)$.

Theorem

Any k -inducible distribution is a mean-preserving contraction of a distribution k -induced by an extreme RCRS.

$$\mathcal{G} = \bigcup_{H \in \mathcal{H}} \text{MPC}(H)$$

where $\mathcal{H}$ is the set of distributions k -induced by an extreme RCRS.